W-S-890B

W-S-890b
December 31, 1963

SUPERSEDING
Int. Fed. Spec. W-S-00890a (GSA-FSS)
December 18, 1961 and
Fed. Spec. W-S-890
October 6, 1944

FEDERAL SPECIFICATION

SWITCH, SNAP, MISCELLANEOUS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers manually operable general-use snap switches for service on alternating current or direct current for which the load ratings do not exceed 60 amperes at 250 volts or less, or 30 amperes at 600 volts or less.

1.2 Classification.

1.2.1 *Types and classes.* Switches furnished under this specification shall be of the following types and classes, as specified (see 6.1).

Type I—Surface
- Class 1—Tumbler
- Class 2—Rotary
- Class 3—Pull
- Class 4—Push

Type II—Junction box (switch mounted on box cover)
- Class 1—Tumbler
- Class 3—Pull
  - Kind C—Ceiling mounting
  - Kind W—Wall mounting

Type III—Through-cord
- Class 1—Tumbler
- Class 4—Push

Type IV—Pendent
- Class 1—Tumbler
- Class 4—Push

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

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Military Standard:

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

Underwriters' Laboratories, Inc., Publication:

UL-20—Standard for Snap Switches.

(Copies may be obtained from Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, N.Y., 11749; 207 E. Ohio Street, Chicago, Illinois, 60611; and 1655 Scott Blvd., Santa Clara, California 95050.)

3. REQUIREMENTS

3.1 Switches covered by this specification shall conform to the applicable requirements of Underwriters' Laboratories, Inc., Standard No. 20 for Snap Switches.

3.2 Fire and Casualty hazards.

3.2.1 Each bidder shall submit to the contracting agency, proof that the switches he proposes to supply under this specification conform to the applicable requirements of Underwriters' Laboratories, Inc., Standard No. 20 for Snap Switches. The label or listing with reexamination of the Underwriters' Laboratories, Inc., may be accepted as evidence that the switches conform to these requirements.

3.2.2 In lieu of the label or listing with reexamination, the bidder may submit independent proof satisfactory to the contracting agency that his switches conform to the published standards including methods of tests of the Underwriters' Laboratories, Inc., Standard No. 20 for Snap Switches.

3.2.3 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of his material.

3.3 General.

3.3.1 Each switch shall be furnished completely assembled and ready for installation and shall not require adjustment or lubrication.

3.3.2 The fit and keying of parts shall be such that disassembly and reassembly for wiring and mounting are simply and easily accomplished by use of a screwdriver. Upon reassembly, the parts shall fit together only in that manner which restores them to their original assembled positions.

3.3.3 Toggle and rotary switches of single-pole, double-pole, or single-heat design shall be plainly marked to show the "off" and "on" positions. Three-heat switches shall be plainly marked to show "off", "high", "medium," and "low" positions, in that order in the clockwise direction.

3.3.4 Unless otherwise specified, any color shall be acceptable for the exposed parts of a switch.

3.4 Ratings.

3.4.1 A switch having a higher rating than that specified in the invitation for bids shall be acceptable in lieu of the switch specified; for example, a switch having a rating of 6 amperes at 125 volts and 3 amperes at 250 volts shall be acceptable in lieu of a switch having a rating of 5 amperes at 125 volts and 3 amperes at 250 volts.

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3.5 Surface switches (type I).

3.5.1 Bases for class 1, 2, or 3 switches shall be either solid or slotted, as specified.

3.5.2 When required, class 3 switches shall be arranged for either ceiling or wall mounting, as specified.

3.5.3 Class 2 heater switches shall be operable both clockwise and counter-clockwise; other class 2 switches, if not operable in both directions, shall turn clockwise.

3.6 Outlet box switches (type 2).

3.6.1 Each switch shall consist of a switch body securely fixed to an outlet box cover and shall be of such design that it will satisfactorily fit a standard outlet box of either 3-1/4- or 4-inch size, as specified.

3.6.2 The wiring terminals shall be so located or protected that when the switch is installed, the terminals will not be forced against the wiring in the box. The clearance between the current-carrying parts and the metal of the outlet box on which the switch is to be mounted shall be not less than 1/8 inch.

3.6.3 The outlet box cover shall be a disk and shall be either metal, a thermosetting plastic approved by UL for the purpose, or porcelain, as specified. The cover shall be of a suitable size and design to properly fit, attach to, and cover a standard outlet box.

3.7 Through-cord and pendent switches (types III and IV).

3.7.1 Unless otherwise specified, cord inlet holes shall be 0.406 inch, plus or minus 1/64 inch, in diameter.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Unless otherwise specified, inspection and test shall be in accordance with Underwriters' Laboratories, Inc., Standard No. 20 for Snap Switches.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of the various levels of packaging and packing shall be as specified in Fed. Std. No. 102.)

5.1 Packaging.

5.1.1 *Level C.* Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level C.* The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required in the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

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5.3.2 *Military agencies.* In addition to any special marking required in the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.
(*b*) Type and class of switch (see 1.2).
(*c*) Voltage and current rating.
(*d*) Number of poles and throws or similar circuit design requirement.
(*e*) Color (see 3.3.4).
(*f*) Type of surface switch base desired (see 3.5.1).
(*g*) Type of mounting desired for class 3 surface switch (see 3.5.2).
(*h*) Size and cover material for outlet box switches (see 3.6.1 and 3.6.3).
(*i*) Diameter of cord inlet holes for types III and IV switches if other than specified (see 3.7.1).
(*j*) Inspection responsibility, if other than specified (see 4.1).

6.2 Transporation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Electric switches, or parts thereof, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:

Electric switches or parts thereof.
Truckload minimum weight 30,000 pounds, subject to Rule 115 National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

User Interest: Army—MO MU

Navy—None

Air Force—None

Copies of this specification may be purchased for 5 cents each.

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